

## ・ NEXUSログイン後のページ

↓ タブの“Lab”を押下。“Launch New Environment ”を押下。

The screenshot shows the Nexus Lab web interface. At the top, there is a navigation bar with tabs: Jobs, Projects, Teams, Backends, Lab (selected), and Docs. A red arrow points to the 'Lab' tab. Below the navigation bar, a dropdown menu is open for the 'Lab' tab, showing options: 'Launch New Environment -> Launch a new coding environment.' and 'Learn More About Nexus Lab Find out more by reading the Nexus docs.' The main content area is divided into three sections. The left section shows user information: Username yasuyoshiyonezawatorquantinuumcom, Org QTM-JP Training, Job Priority 5, User ID e267b ... c3097, and Default Group UTokyo2024Summer. The middle section shows a list of jobs with columns for status, name, and submission time. The right section shows 'Available Backends' and 'Nexus Lab' information.

**Navigation Bar:** Jobs, Projects, Teams, Backends, Lab (selected), Docs

**Search Bar:** Search documentation...

**Lab Tab Dropdown:**

- Launch New Environment -> Launch a new coding environment.
- Learn More About Nexus Lab Find out more by reading the Nexus docs.

**User Information:**

- Username: yasuyoshiyonezawatorquantinuumcom
- Org: QTM-JP Training
- Job Priority: 5
- User ID: e267b ... c3097
- Default Group: UTokyo2024Summer

**Usage Section:**

- Simulation: Quota Resets in 13 Days
- 0 s Used, 0 s Remaining
- Compilation: Quota Resets in 13 Days
- 0 s Used, 0 s Remaining
- Lab: Quota Resets in 13 Days
- 4 h, 5 m Used
- Storage

**Jobs List:**

| Status | Job Name                                          | Submission Time                                                       |
|--------|---------------------------------------------------|-----------------------------------------------------------------------|
| ✓      | compile-job-convex-terracotta-resolute-versipelle | Compile job submitted by yasuyoshi.yonezawa+torq@quantinuum.com 6 か月前 |
| ✓      | execute-job-conical-earthy-perky-laevigata        | Execute job submitted by yasuyoshi.yonezawa+torq@quantinuum.com 6 か月前 |
| ✓      | compile-job-adnexed-dread-imperious-badia         | Compile job submitted by yasuyoshi.yonezawa+torq@quantinuum.com 6 か月前 |
| !      | execute-job-subdecurrent-stump-bewitching-viscosa | Execute job submitted by yasuyoshi.yonezawa+torq@quantinuum.com 6 か月前 |

**Available Backends:**

View quantum hardware, simulator and emulators available to you on your current plan.

[View Backends](#)

**Nexus Lab:**

Online JupyterHub environments with preinstalled packages.

[Launch a Codespace](#)

[View InQuanto Examples](#)

**Nexus Support:**

If you're experiencing issues, please include as much detail as possible. Error messages, steps you have followed, or screenshots are particularly helpful.

[QCSupport@quantinuum.com](mailto:QCSupport@quantinuum.com)

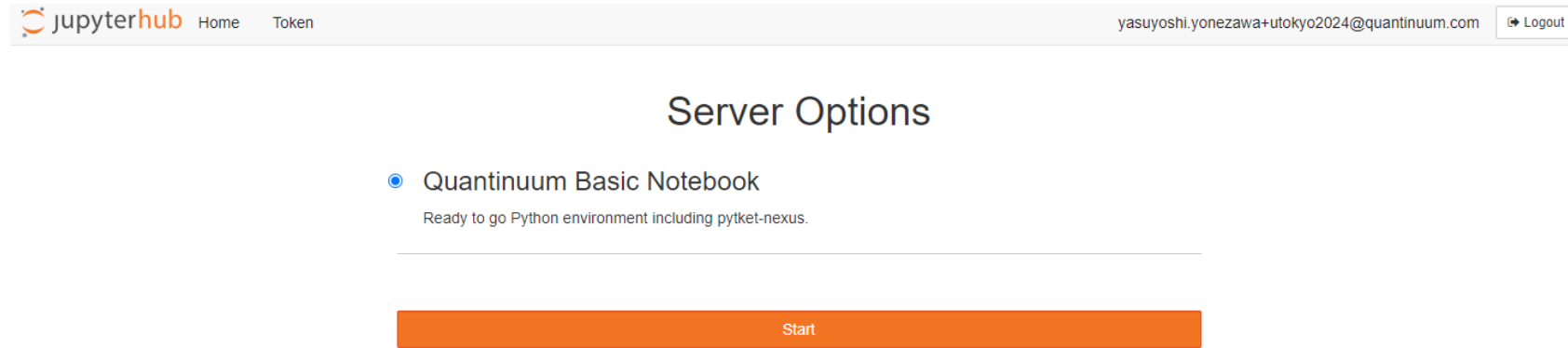
**Documentation:**

Learn how to log in, submit jobs to quantum hardware and view results.

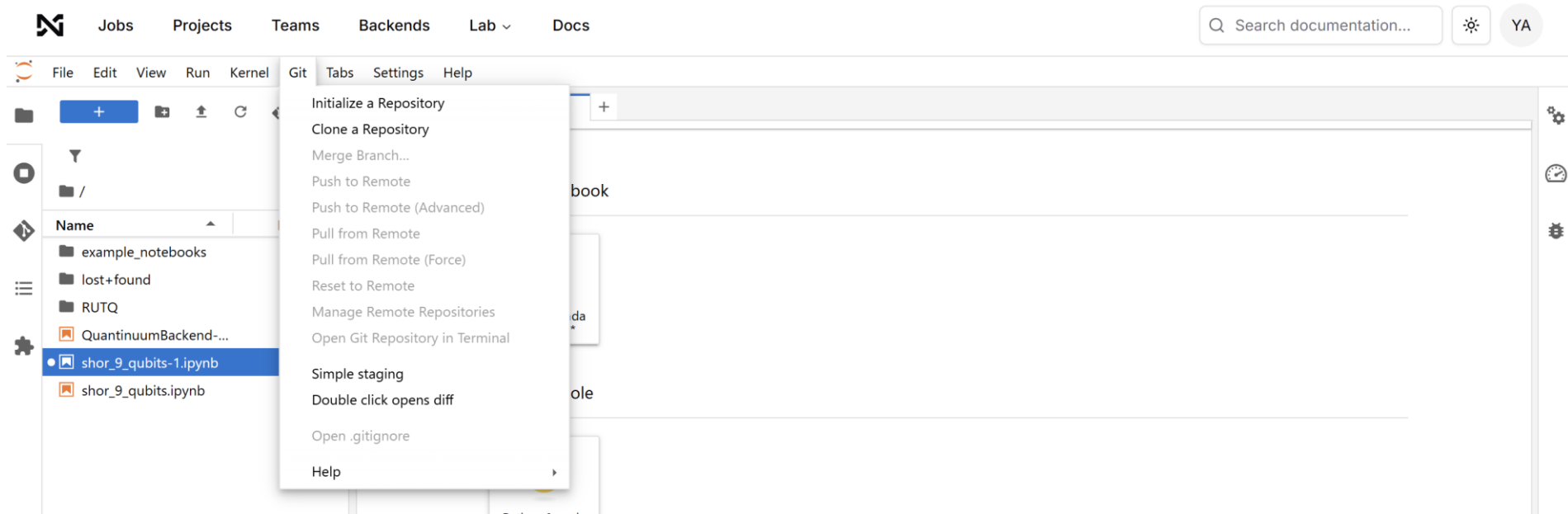
[User Guide](#)

## ▪ Launch New Environment選択後のページ

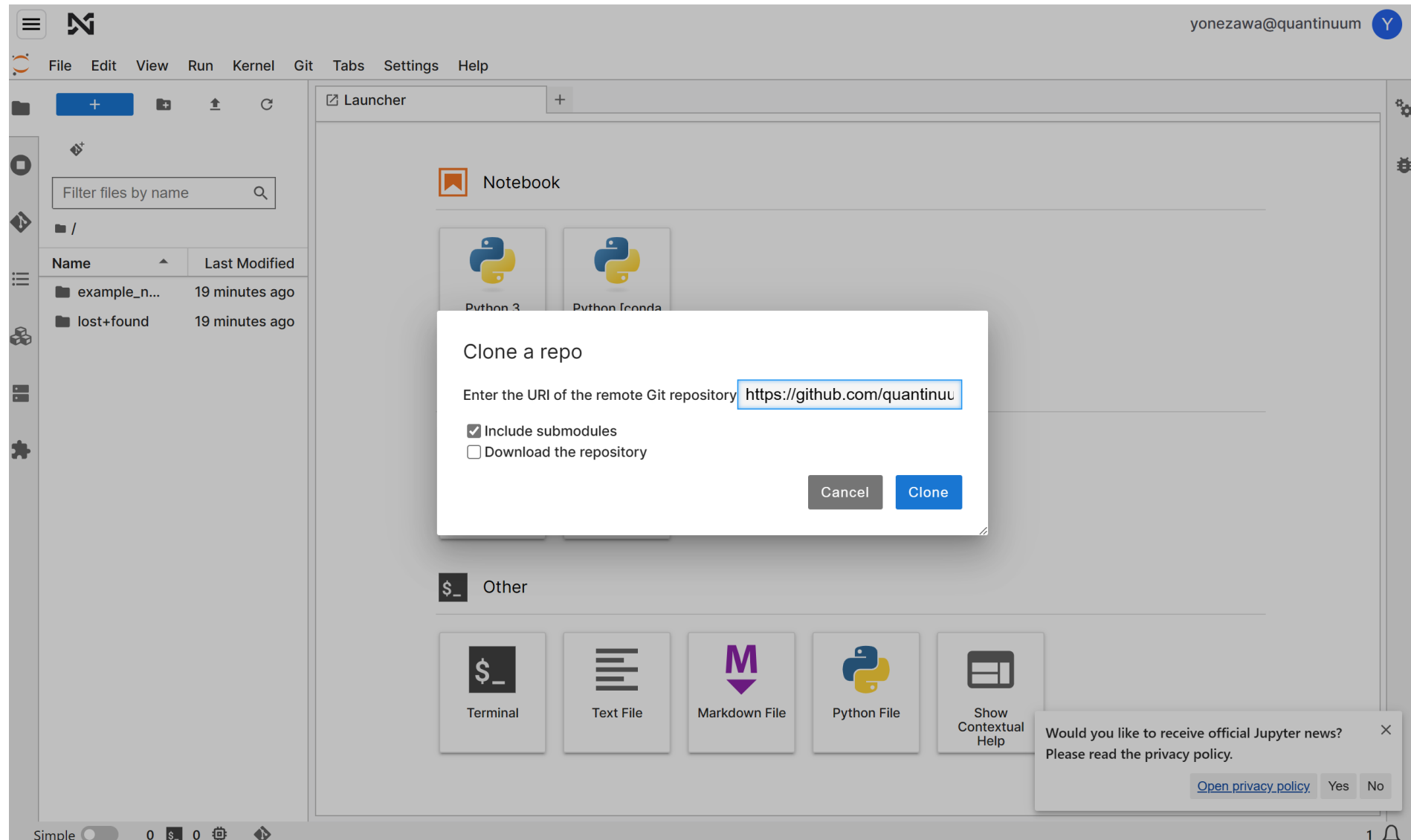
“start”を押下



しばらくすると“JupyterLab”が立ち上がります “Git”タブを押下し、“Clone a Repository”を選択



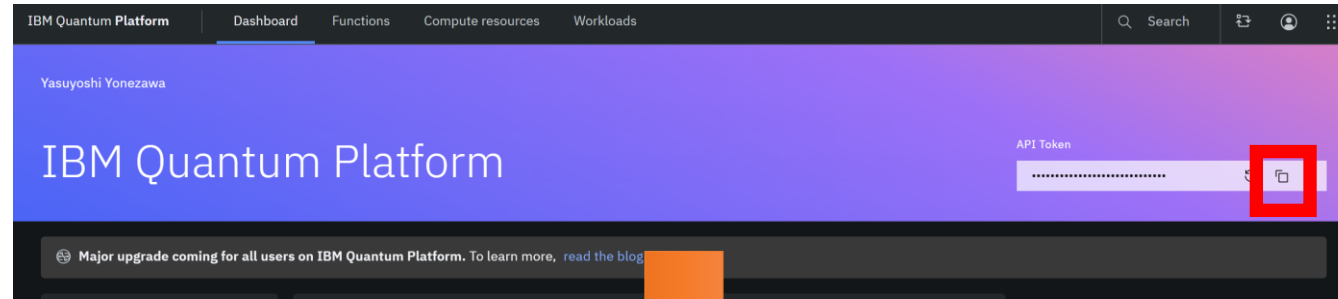
”Clone a Repository”を選択後、「https://github.com/quantinuum-jp/RUTQ.git」を入力し、Cloneを選択するとハンズオンで利用するファイルがインポートされます。



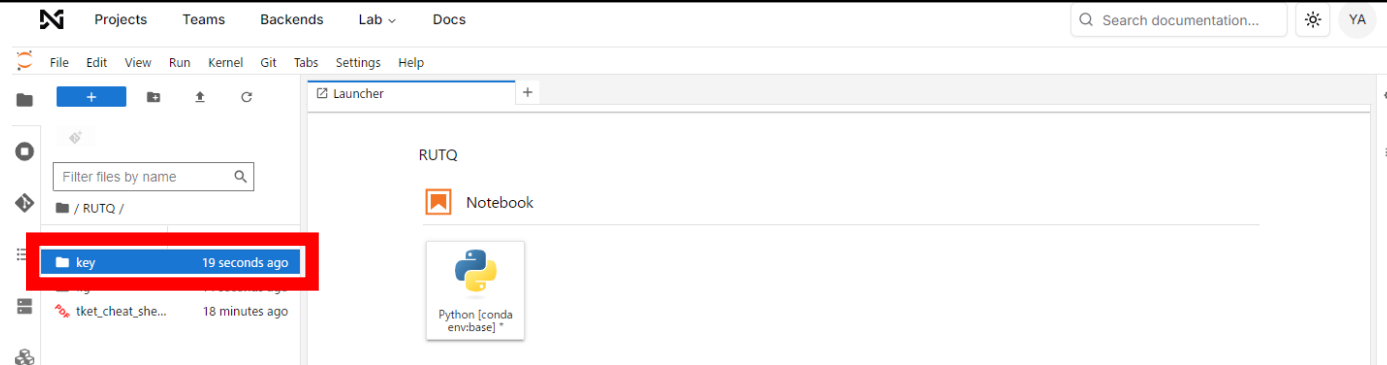
# IBMQ API tokenの準備

<https://quantum.ibm.com/> にアクセス  
(アカウントがなければ、各自が作成する必要があります。無料です)

クリックしてtokenをコピー(ハンズオンで実施)



Keyフォルダーのibm-tokenファイルを開きペースト(ハンズオンで実施)



# 自身のPCにTKETが利用できるPython環境の準備する

1. Python 3.11, 3.12のいずれかを用意ください。

```
pip install pytket
```

でTKETが利用できるpythonライブラリーpytketがインストールできる。

2. pytket extension関連ライブラリーのインストールも忘れずに。

```
pip install pytket-quantinuum
```

```
pip install pytket-quantinuum[pecos]
```

```
pip install pytket-qiskit
```

```
pip install pylatexenc
```

```
pip install matplotlib
```